

LESSON 2-2 • SESSION 1 • PRACTICE • FORM B**Building a Histogram**

Learning target – I can group data into equal intervals and build a histogram from a frequency table. • 6.SP.B.4

REMEMBER

- An **interval** is a group of numbers, like 10–19. Intervals are equal in width and **never overlap**.
- **Frequency** = how many values fall in an interval. It sets the **bar height**.
- Histogram bars **touch**. All the bar heights add up to the number of values.

WORKED EXAMPLE

Walk times (min): 4, 15, 9, 22, 11, 18, 27. Use 0–9, 10–19, 20–29.

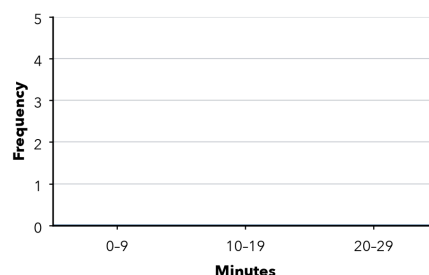
Step 1 0–9 holds 4 and 9: frequency 2.**Step 2** 10–19 holds 15, 11, 18: frequency 3. 20–29 holds 22, 27: frequency 2.**Step 3** Draw touching bars of height 2, 3, 2.**Answer Check:** $2 + 3 + 2 = 7$ values**WORDS** histogram (*histograma*) • interval (*intervalo*) • frequency (*frecuencia*)**PRACTICE**

show your work • label units

- 1. Route 4.** Nine students timed their bus rides (min): 5, 14, 22, 11, 8, 17, 26, 3, 19. Complete the table. Then draw the histogram.

Hint Cross off each value as you tally it.

Interval	Frequency
0–9	
10–19	
20–29	

**Fewest rides** _____

- 2. Pushups.** 9 students did these pushups: 17, 28, 6, 21, 14, 25, 29, 23, 12. Find each frequency and the total.

a. 0–9: ____ **b.** 10–19: ____ **c.** 20–29: ____ **d.** Total: ____

- 3. Read the Table.** 30–39: 6, 40–49: 9, 50–59: 5, 60–69: 2.

Hint Add the frequencies; the biggest one is the tallest bar.**a.** Values in all: ____**b.** Tallest bar: ____

- 4. Missing Bar.** A histogram shows 12 values. Two bars have heights 3 and 5. How tall is the third bar?

Hint All the bar heights add up to the total.

Show your work.

Height _____

- 5. Car Wash.** Cars washed on 8 Saturdays: 15, 31, 8, 26, 37, 19, 22, 34. Find each frequency.

Hint Put the counts in order first.

a. 0–9: ____ **b.** 10–19: ____ **c.** 20–29: ____ **d.** 30–39: ____

- 6. Overlapping Bins.** Desmond sorts 3, 8, 12, 15, 19 into the intervals 0-5, 5-10, 10-15, 15-20. What is wrong? Write intervals that work.
- Hint** Look for a number that fits in two intervals.

- 7. Choose the Intervals.** Plant heights run from 42 to 87 cm. Write five equal intervals of width 10 that cover every height.
- Intervals** _____

- 8. Field Day.** Twelve runners finished in 31, 44, 38, 52, 47, 35, 41, 56, 39, 48, 33, 45 seconds. Complete the table.
- a.** Which interval holds the fewest runners? _____
- b.** Why does a histogram show the times better than the list?

Interval	Frequency
30-39	
40-49	
50-59	

FIND THE ERROR explain the mistake, then fix it

- 9. Nadia's Work.** For the values 4, 12, 18, 26, Nadia drew bars over 0-9, 10-19 and 20-29 with a space between each bar. Explain the mistake and describe the correct histogram.
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TEST PRACTICE choose the best answer

- 10.** Which set of intervals is correct for a histogram?
- A** 0-5, 5-10, 10-15 **B** 0-9, 10-14, 15-29
- C** 0-9, 10-19, 20-29 **D** 0-10, 10-20, 20-30
- 11.** Histogram bars touch because the intervals –
- A** are different widths **B** leave no gaps between them
- C** overlap **D** have the same frequency

SELF-CHECK CODE find your answer · copy its letter into that problem's box · spell a math word

50-59 D	5 O	22 N	30-39 X	20-29 C	12 Q	9 U	40-49 T	4 E	8 Z
	#1	#2c	#2d	#3a	#3b	#4	#8a		

CHECK YOURSELF	☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____
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